

CS5239 Computer System Performance Analysis

L2: Methodologies and Capacity Planning

Outline

- Terminology
- Perspectives in system's performance
- Methodologies
- Capacity planning
 - What?
 - How?

Learning outcomes

By the end of this lecture, you should be able to

Characterise

workload intensity, mix, variability, and growth

Plan

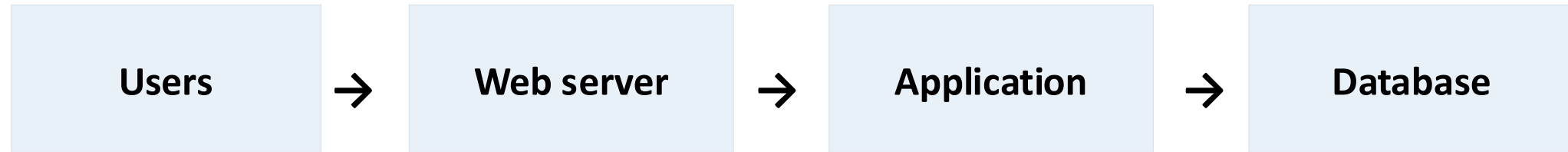
capacity using service levels, models, forecasts, and cost

Course connection

- L7 provides practical observation tools later
- L8 analyses resource limits and demand
- L9 monitors growth and service levels
- L11 validates workloads with benchmarks

A running case for L2

A virtual car dealership must scale its search and purchase service



Workload

70% document and image retrieval
25% database search
5% requests become purchases

Service level

Search response time below 4 seconds
High availability during sales
campaigns

Decision

When should capacity be added
and what configuration is cost
effective

Terminology for decisions

Latency

Time an operation spends waiting or the delay before service begins

Response time

Time from request to completed response including service and transfer

Throughput

Rate of work completed such as requests per second or IOPS

Utilisation

Fraction of time or capacity used during an observation interval

Saturation

Work that cannot be served immediately and begins to queue

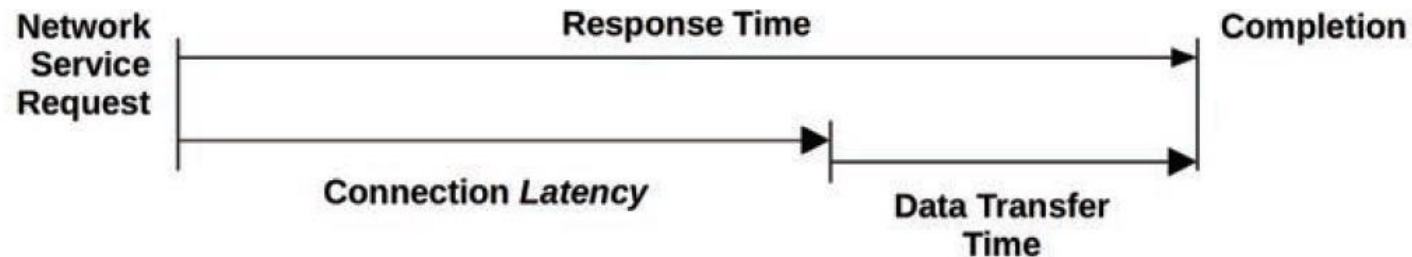
Bottleneck

Resource or interaction that limits system performance

Use the term that matches the decision: user delay, completed work, resource pressure, or limiting factor

Terminology (1)

- **Latency**: A measure of time an operation spends waiting to be serviced. In some contexts, it can refer to the entire time for an operation, equivalent to response time
- **Response time**: The time for an operation to complete. This includes any time spent waiting and time spent being serviced (service time), including the time to transfer the result.



Terminology (2)

- **Throughput**: The rate of work performed. Especially in communications, the term is used to refer to the data rate (bytes per second or bits per second). In some contexts (e.g., databases) throughput can refer to the operation rate (operations per second or transactions per second).
- **IOPS**: Input/output operations per second is a measure of the rate of data transfer operations. For disk I/O, IOPS refers to reads and writes per second.
- **Workload**: The input to the system or the load applied is the workload. For a database, the workload consists of the database queries and commands sent by the clients.

Time scales

- System components operate over vastly different time scales (orders of magnitude)

Event	Latency	Scaled
1 CPU cycle	0.3 ns	1 s
Level 1 cache access	0.9 ns	3 s
Level 2 cache access	3 ns	10 s
Level 3 cache access	10 ns	33 s
Main memory access (DRAM, from CPU)	100 ns	6 min
Solid-state disk I/O (flash memory)	10–100 µs	9–90 hours
Rotational disk I/O	1–10 ms	1–12 months
Internet: San Francisco to New York	40 ms	4 years
Internet: San Francisco to United Kingdom	81 ms	8 years
Lightweight hardware virtualization boot	100 ms	11 years
Internet: San Francisco to Australia	183 ms	19 years
OS virtualization system boot	< 1 s	105 years
TCP timer-based retransmit	1–3 s	105–317 years
SCSI command time-out	30 s	3 millennia
Hardware (HW) virtualization system boot	40 s	4 millennia
Physical system reboot	5 m	32 millennia

Terminology (2)

- **Utilization**: For resources that service requests, utilization is a measure of how busy a resource is, based on how much time in a given interval it was actively performing work. For resources that provide storage, utilization may refer to the capacity that is consumed (e.g., memory utilization).
- **Saturation**: The degree to which a resource has queued work it cannot service.
- **Bottleneck**: In systems performance, a bottleneck is a resource that limits the performance of the system. Identifying and removing systemic bottlenecks is a key activity of systems performance.

Utilization

Time-based

- The average amount of time the server or resource was busy
 - $U = B/T$
- Where:
 - U = utilization
 - B = total busy during T
 - T = the observation period
- Commonly measured for devices in a computer system
- Example: A CPU is busy for 7 seconds during a 10-second interval

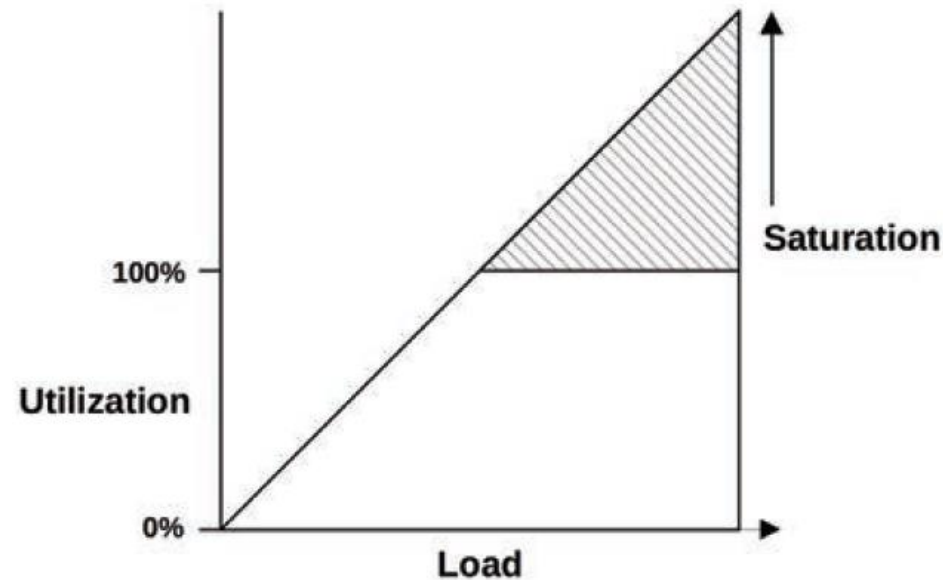
Capacity-based

- A system or component (such as a disk drive) is able to deliver a certain amount of throughput and the utilization is the proportion the system or component is working at
- Example: A database can sustainably process 2,000 queries per second and currently receives 1,200 queries per second

A disk may have 100% time-based utilisation because it is continuously servicing requests, while achieving only 60% of its maximum possible throughput because the requests are small or random

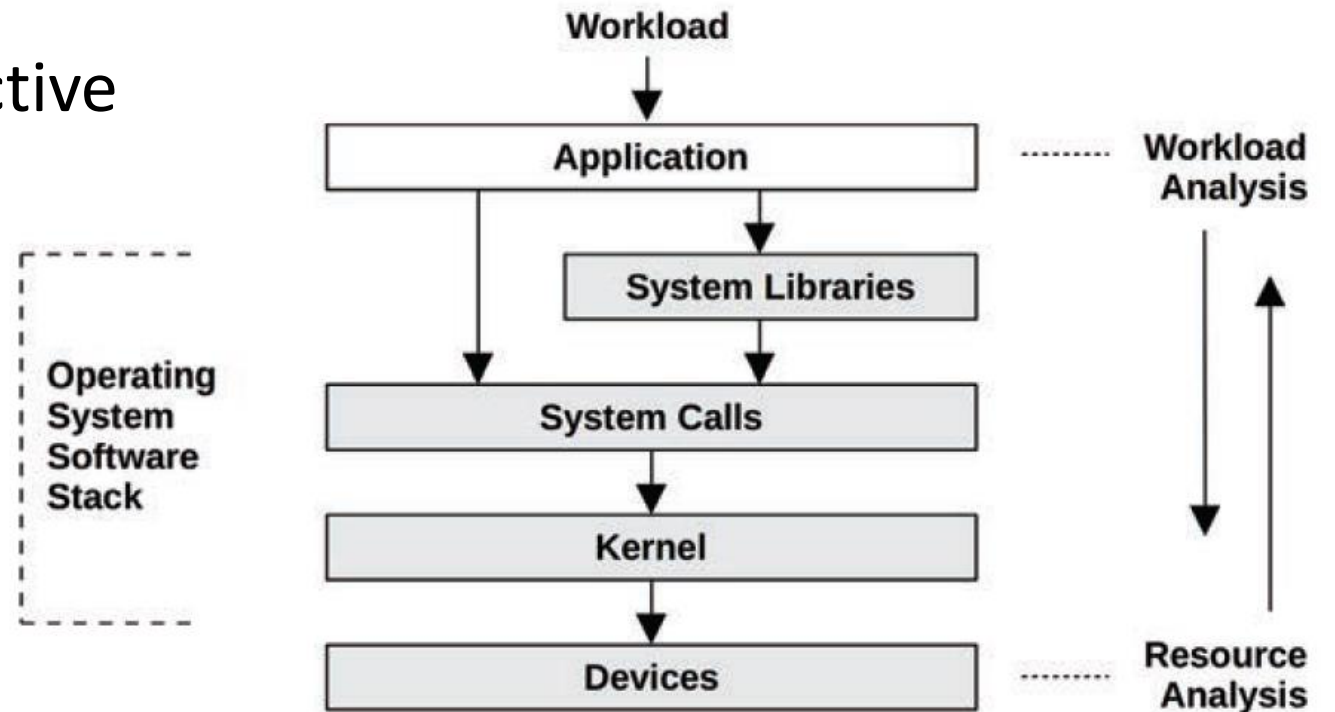
Saturation

- The degree to which more work is requested of a resource than it can process
- Saturation begins to occur at 100% utilization (capacity-based)
 - Extra work cannot be processed and begins to queue



Perspectives

- The resource analysis perspective
→ system administrators
- The workload analysis perspective
→ application developers



Resource Analysis

- Begins with analysis of the system resources: CPUs, memory, disks, network interfaces, buses, and interconnects
- Includes
 - Performance issue investigations
 - Capacity planning
- Focuses on utilization
 - Demand - supply

Workload Analysis

- Examines the workload applied and how the application is responding
- Targets
 - Requests: The workload applied
 - Latency: The response time of the application
 - Completion: Looking for errors
- Metrics
 - Throughout
 - Latency

Methodologies

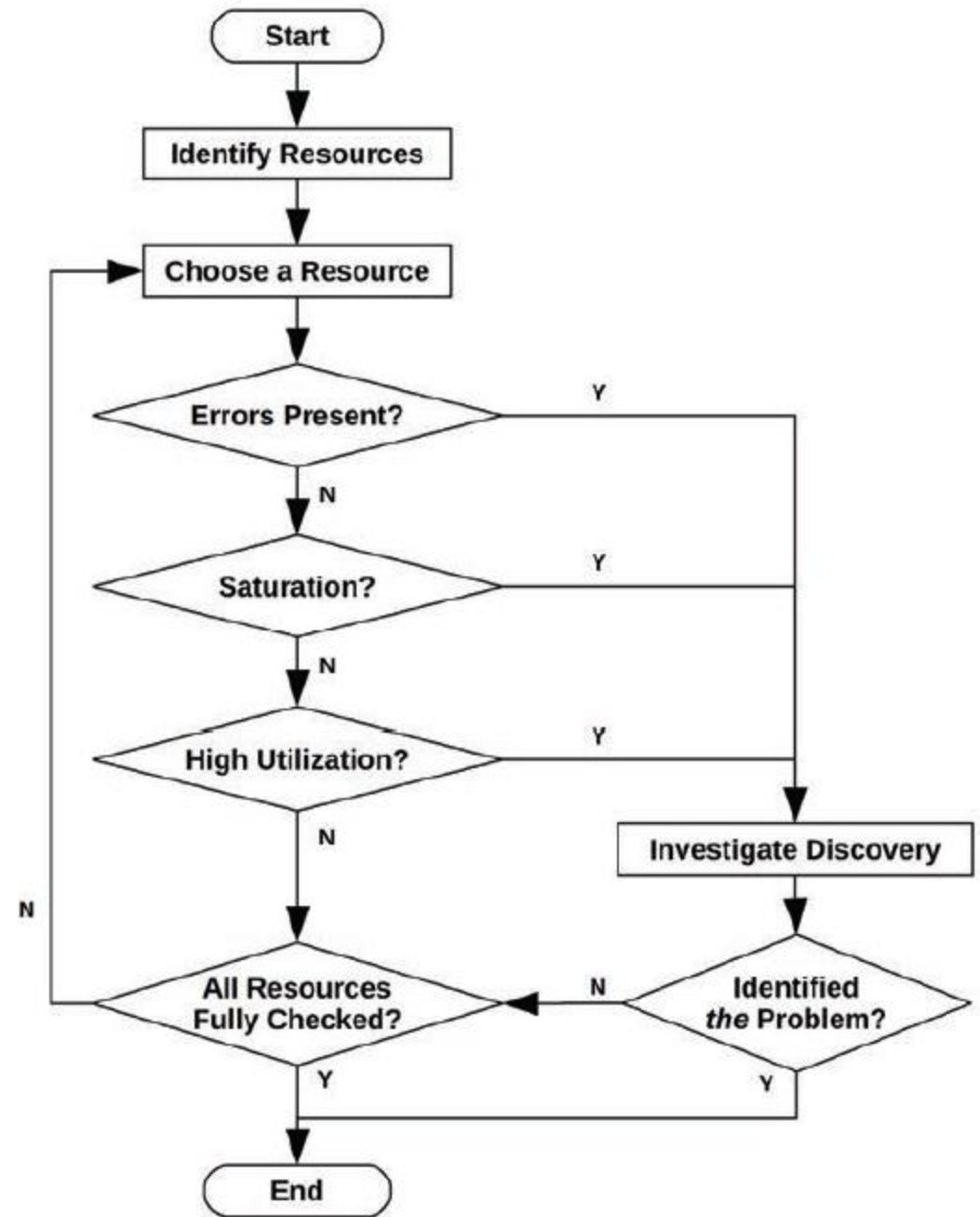
- Know where to begin analysis and how to proceed
- Methods
 - Tools method
 - USE method
 - RED method
 - Workload characterization
 - Monitoring
 - Capacity planning [this lecture]
 - Modelling [next lecture]

Tools Method

1. List available performance tools (optionally, install or purchase more).
2. For each tool, list useful metrics it provides.
3. For each metric, list possible ways to interpret it.

The USE Method

- For every resource, check
 - Utilization
 - Saturation
 - Errors
 - Resources:
 - CPUs
 - Main memory
 - Network interfaces
 - Storage devices
 - Accelerators
- machine health



The RED Method

- For every service (cloud services in a microservice architecture), check
 - Request rate
 - Errors
 - Duration
- user health

Workload Characterization

- Focuses on the **input** to the system, rather than the resulting performance
- Workload questions
 - **Who** is causing the load? Process ID, user ID, remote IP address?
 - **Why** is the load being called? Code path, stack trace?
 - **What** are the load characteristics? IOPS, throughput, direction (read/write), type? Include variance (standard deviation) where appropriate.
 - **How** is the load changing over time? Is there a daily pattern?

Monitoring

- Records performance statistics over time (a time series) so that the past can be compared to the present and time-based usage patterns can be identified
- Useful for
 - Capacity planning
 - Quantifying growth
 - Showing peak usage
- Historic values can also provide context for understanding the current value of performance metrics
 - Time-based patterns

Methodologies

- Know where to begin analysis and how to proceed
- Methods
 - Tools method
 - USE method
 - RED method
 - Workload characterization
 - Performance monitoring
 - Capacity planning [this lecture]
 - Modeling [next lecture]

Capacity Planning

Capacity planning

- Examines how well the system will handle load and how it will scale as load scales
- Study resource limits and factor analysis, and modeling
- Process to determine and fulfill future workload demands of an organization
 - “predicts when the service levels will be violated as workload evolves and determine cost-effective way of delaying system saturation”
- Challenges: usage load (resource demand) fluctuations, peak usage, cost of provisioning, ...

Example

Insufficient system capacity cause of StarHub's broadband problem

- Apr 22, 2017

Penalty: \$15K to \$270K fine for every 30 mins of outage

Insufficient system capacity cause of StarHub's broadband problem last October

Disruptions were not due to cyber attack

IRENE THAM, THE STRAITS TIMES

StarHub was warned by the authorities yesterday for two disruptions to its home broadband network in October last year.

Investigation findings now show that the disruptions were not due to direct cyber attacks as the telco had suspected.

Findings released yesterday by the Infocomm Media Development Authority (IMDA) and the Cyber Security Agency of Singapore (CSA) showed that StarHub did not have sufficient system capacity to cope with an Internet traffic surge.

They added that the telco has since increased its capacity.

IMDA also directed StarHub to do an independent review of its infrastructure, saying it "will not hesitate to take sterner action should a similar incident happen in future".

IMDA's code spells out the minimum requirements for service availability. Breaching these requirements — such as a loss of 70 per cent or more of a telco's Internet bandwidth — may result in a fine ranging from

"(IMDA) will not hesitate to take sterner action should a similar incident happen in future"

— IMDA's warning to StarHub

\$15,000 to \$270,000 for every 30 minutes of outage.

Subscribers could not surf the Internet intermittently for up to two hours each time on Oct 22 and 24 as traffic spiked on StarHub's DNS, which maps Web addresses to a machine-readable string of numbers to connect users to websites.

When the DNS is not operating optimally, users may not be able to access the websites.

The two disruptions came on the heels of a cyber attack the

week before on US-based DNS service provider Dyn.

That was why IMDA and CSA did not rule out the possibility of a cyber attack.

The surge on StarHub's network was indirectly linked to Dyn, said IMDA and CSA.

Affected content providers in the US were routed to a different technology that conducts extra verification checks on traffic, including those from StarHub broadband users.

The extra checks generated more traffic that clogged StarHub's network, resulting in surfing slowdowns.

This could have prompted broadband users here to repeatedly refresh their Web browsers, which added to the jam.

Mr Chong Siew Loong, StarHub's chief technology officer, said: "We assure our customers and the regulator that we will continuously review our security posture and enhance network resilience in partnership with network and security providers."

itham@sph.com.sg

Service outages

- Examples:
 - Gmail: Feb 2011 – 120,000 of 150m users without email over the weekend
 - Paypal: Nov 2010 – offline for 4.5 hours (network equipment failure)
 - Facebook: Sep 2010 – offline for 2.5 hours (database overload)
 - Microsoft Azure Cloud: Mar 2008 – down for 22 hours
 - Dropbox: Feb 2013 – down for 48 hours (200m users)
 - M1 3G services: Jan 2013 – down for 69 hours
 - DBS Online Banking: Dec 2015 – down for 3 hours
- Service outages result in loss of revenue and fines

Service outages

The inverse of system outage is

$$\text{availability (\%)} = \frac{\text{total system uptime}}{\text{total observation period}} \times 100$$

Examples

- Is 99% availability good?
- On-premise system: 99.999% (gold standard)
 - What is the downtime?

Example (1)

- A large online vehicle marketplace (server) connects shoppers with thousands of dealers and inventory feeds
- Users access the service through a web and mobile application to
 - browse vehicle listings, photos, and specifications
 - filter and search inventory by make, model, location, price, and availability
 - submit enquiries, request a test drive, or start a purchase application
- **The platform receives three main types of workload:**
 - read-heavy catalogue and image requests
 - database search and filtering requests
 - write and transaction requests such as enquiries and purchases

Example (2) – What if?

- Management wants to answer the following questions:
 - Will the server support the load increase while keeping response time for each request (T), below 4 seconds?
 - If not, at which point will its capacity be saturated and why?
 - How much money could be lost daily (revenue loss) if the server saturates when the load increases?

Example (3)

- Let T denotes the response time for search transaction:
 - if $T \leq 4$ seconds then user will successfully complete search request
 - if $4 < T \leq 6$ seconds then 60% of the users will abort the search
 - and if $T > 6$ seconds then 95% of the users will abort the search
- Assume that 5% of search transactions generate a car sale, and a revenue of \$18,000 on average

Example (4): Performance results

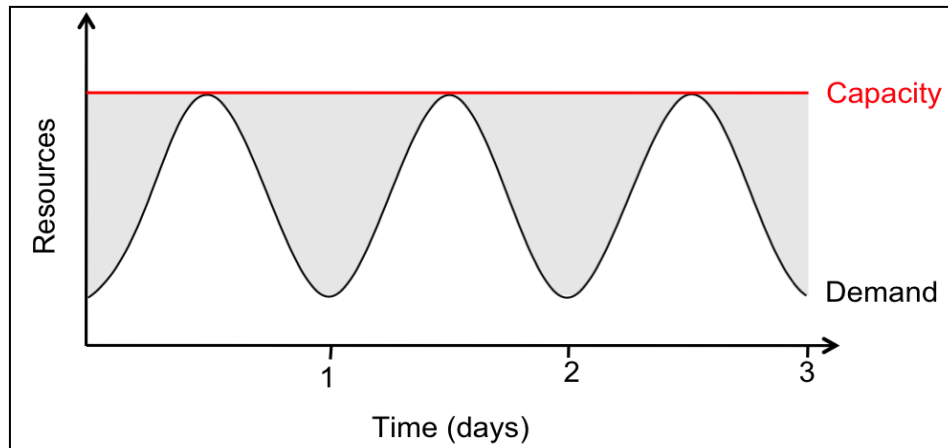
Consequences:

- user dissatisfaction
- loss of revenue
- productivity decreases
- external image of company

	Workload			
	Current			
Search per day	92,448			
Response time (sec)	2.86			
Sales per day	4,622			
Sales lost (%)	0			
Daily revenue	83,203			
Potential daily revenue	83,203			
Daily revenue lost	NA			

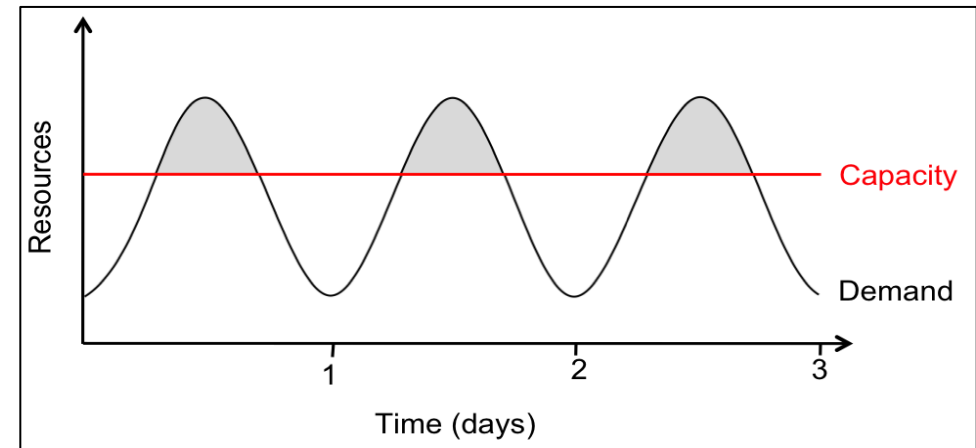
- Revenue in \$1,000

Resource Provisioning & Revenue



a. Provisioning for Peak Load

Even if peak load can be correctly anticipated, **resources are wasted** (shaded area) during non-peak times



b. Under-provisioning

loss of potential revenue from users not served (shaded area)

Reference: Above the Clouds: A Berkeley View of Cloud Computing, 2009.

Provisioning strategies

Lead strategy

Add capacity before demand arrives

Useful for fixed launch events, campaigns, or registration windows

Risk: idle resources if the forecast is wrong

Lag strategy

Add capacity after resources approach their limit

Useful when capacity is expensive and demand is uncertain

Risk: users experience degradation first

Match strategy

Add capacity in small increments as demand grows

Useful for gradual growth and elastic infrastructure

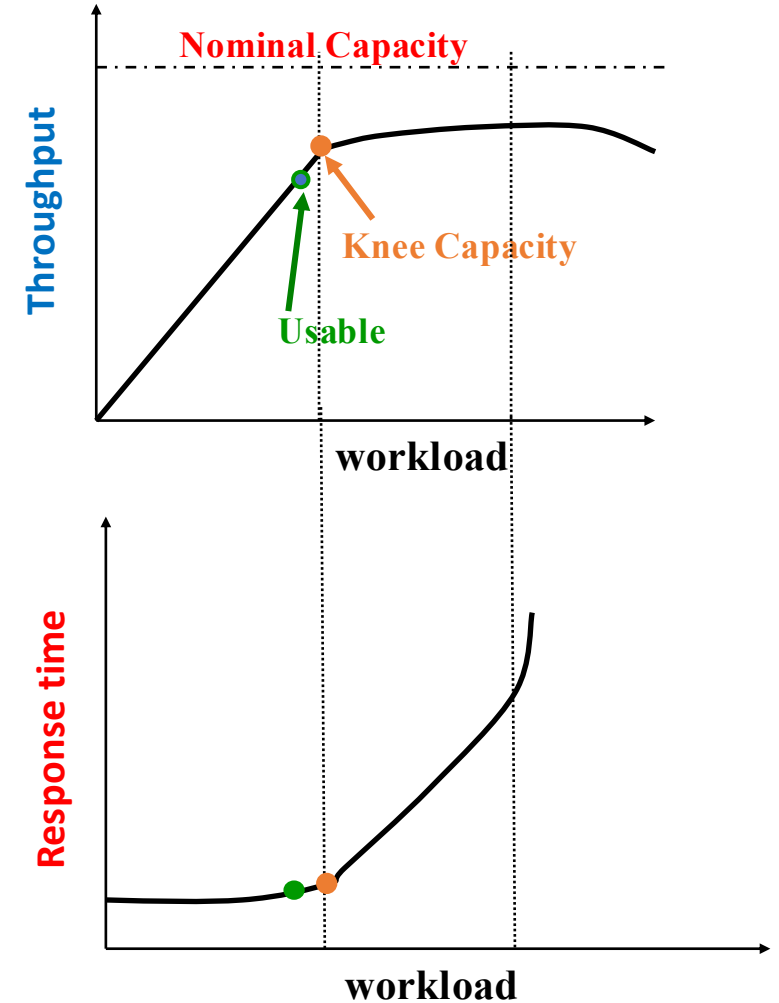
Risk: scaling delay during bursts

Choose the strategy using demand variability, scaling delay, SLA risk, and cost

Capacity of a System

- **Nominal capacity:** maximum throughput under **ideal** workload conditions, e.g. bandwidth (in bps) of a network
- **Usable capacity:** the highest throughput that still satisfies the specified response-time limit
- **Knee capacity:** the workload level beyond which response time increases rapidly while throughput increases little or stops increasing

Note: inverse relationship between **throughput** (management) and **response time** (users)



Service-Level Agreements (SLA)

- Determines what a user of an application can expect in terms of response time, throughput, system availability, reliability, ...
 - Focus on metrics that users can understand
 - Set easy-to-measure goals
 - Tie IT costs to your SLAs

Service Level Agreements: Examples

- Response time for trivial database queries should not exceed 2 seconds
- Web services: 99% availability and less than 1-sec response time for 90% of the HTTP requests for small documents

Capacity planning

Capacity planning predicts whether future demand can be served within agreed limits

Inputs

- Workload growth
- Resource limits
- System configuration
- Service levels
- Cost constraints

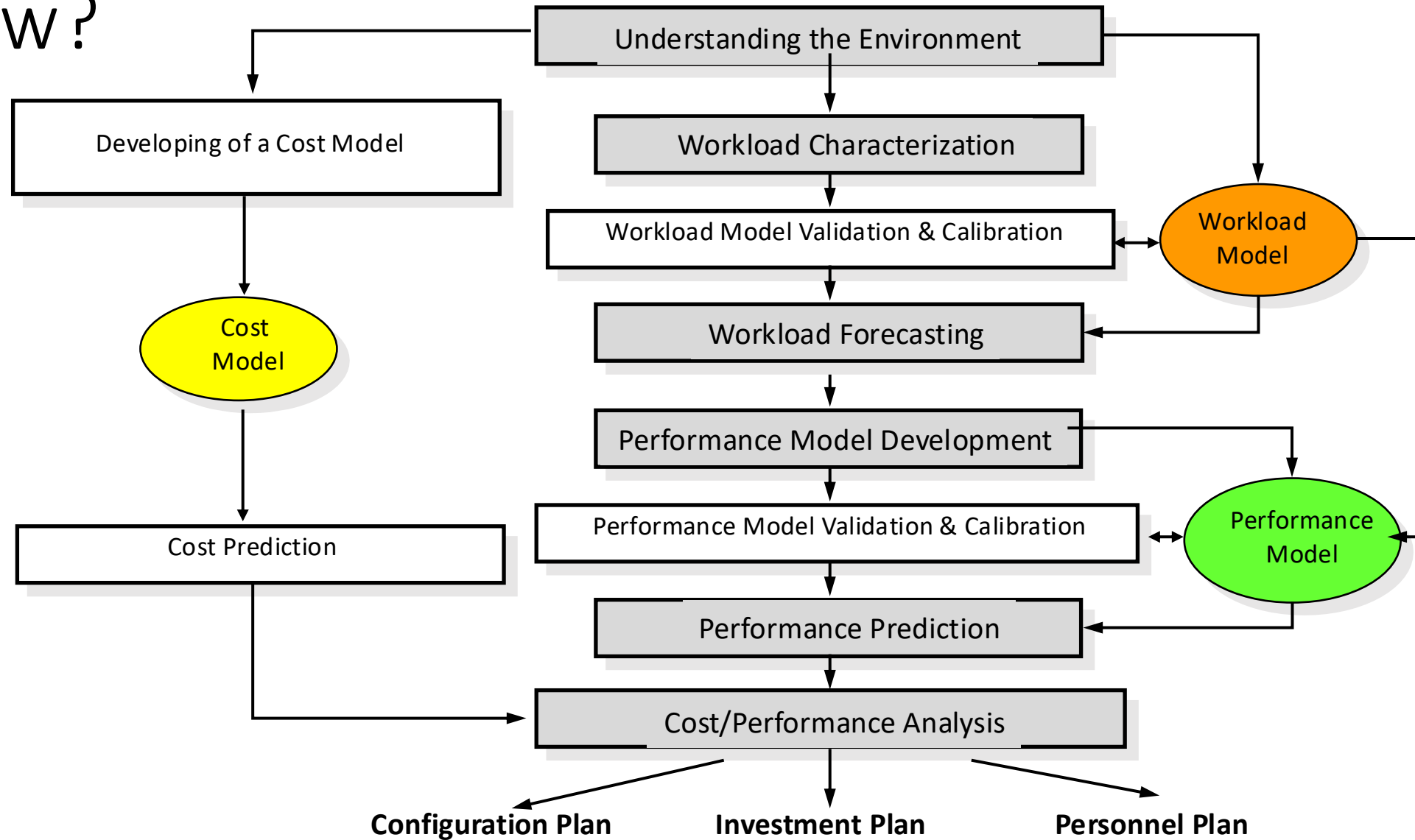
Outputs

- Saturation point
- Expected response time
- Required capacity
- Cost-effective alternative
- Provisioning date

Risks

- Peaks and bursts
- Forecast error
- Provisioning delay
- Changing workload mix
- Model uncertainty

How?



Three models support one capacity decision

Workload, performance, and cost models answer different parts of the same question

Workload model

- Request mix
- Intensity
- Resource demand
- Forecast growth

Performance model

- Response time
- Throughput
- Utilisation
- Queue length
- Saturation point

Cost model

- Hardware and software
- Network and electricity
- Support and staffing
- Startup and operating cost

Choose the least expensive configuration that meets the SLA

3 Models to Support Capacity Planning

- **Workload model**: captures resource demands and workload intensity characteristics of the load for different types of transactions and requests
- **Performance model**: predict response times, utilizations, and throughputs, etc. as a function of the system description and workload parameter
 - These performance metrics are matched with service-level agreements to determine if capacity of the system is adequate.
- **Cost model**: startup/operating costs such as software, hardware, telecommunications, electricity and support expenditures

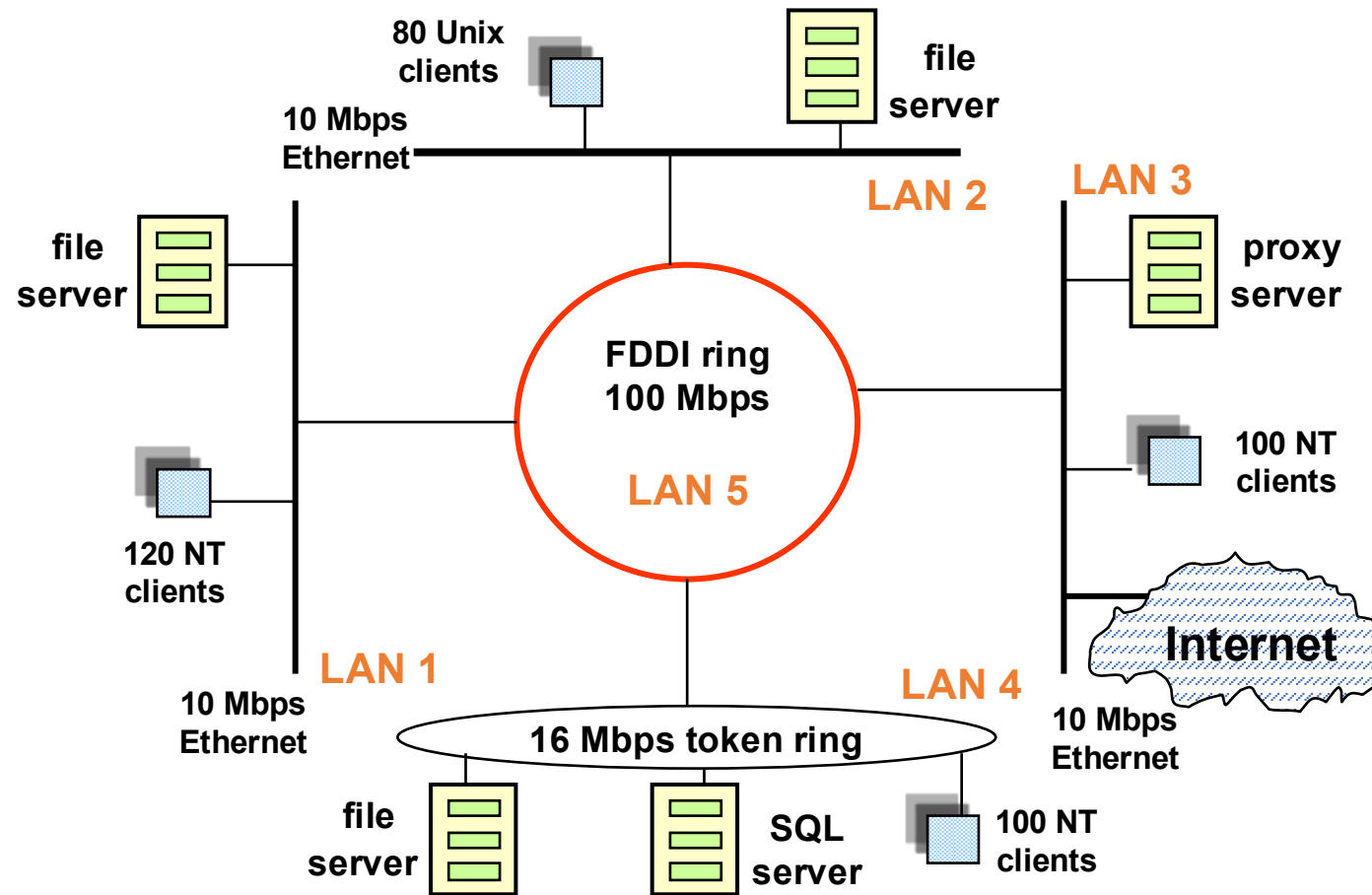
Steps for Capacity Planning

1. Understanding the Environment
2. Workload Characterization & Modeling
3. Workload Forecasting
4. Performance Modeling & Prediction
5. Cost Model

Understanding the Environment

- What kind of
 - hardware (clients and servers)
 - software (OS, middleware, and applications)
 - network connectivity and protocols
- are present in the environment

Example: Understanding the Environment



Elements in Understanding the Environment

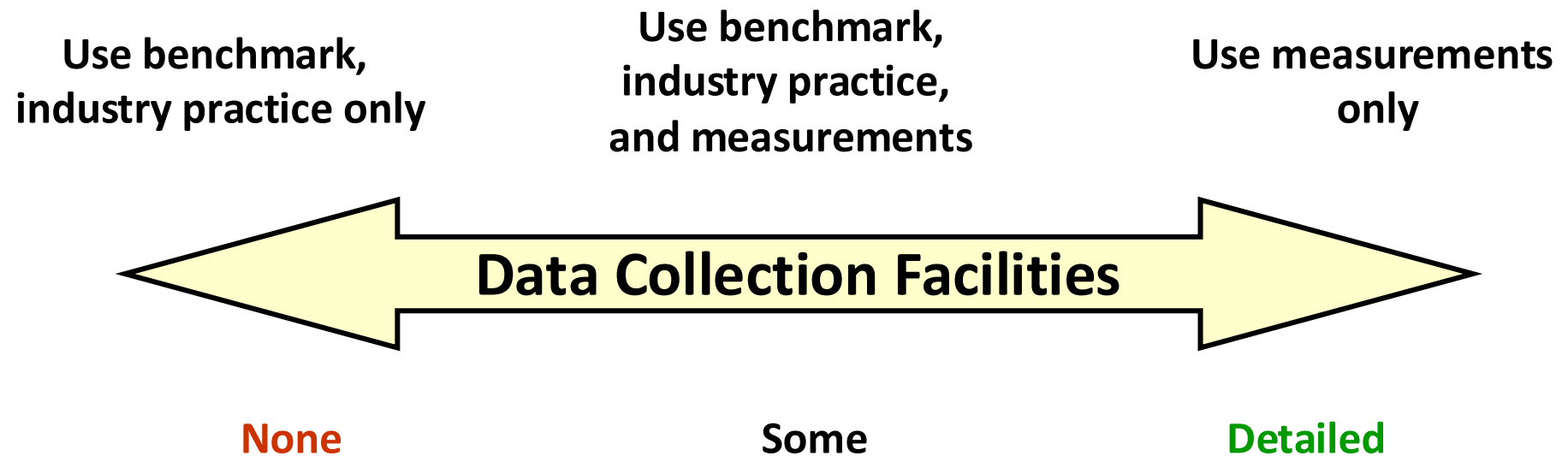
Element	Description
Client platform	Quantity and type
Server platform	Quantity, type, configuration and functions
Middleware	Type (e.g. TP monitors)
DBMS	Type
Application	Main types of applications, criticality, etc.
Network connectivity	Network diagrams with LANs, WANs, routers, servers, etc.
SLAs	Existing SLAs per application
Procurement procedures	Elements of the procurement process, expenditure limits, justification procedures for acquisitions.

Workload Characterization

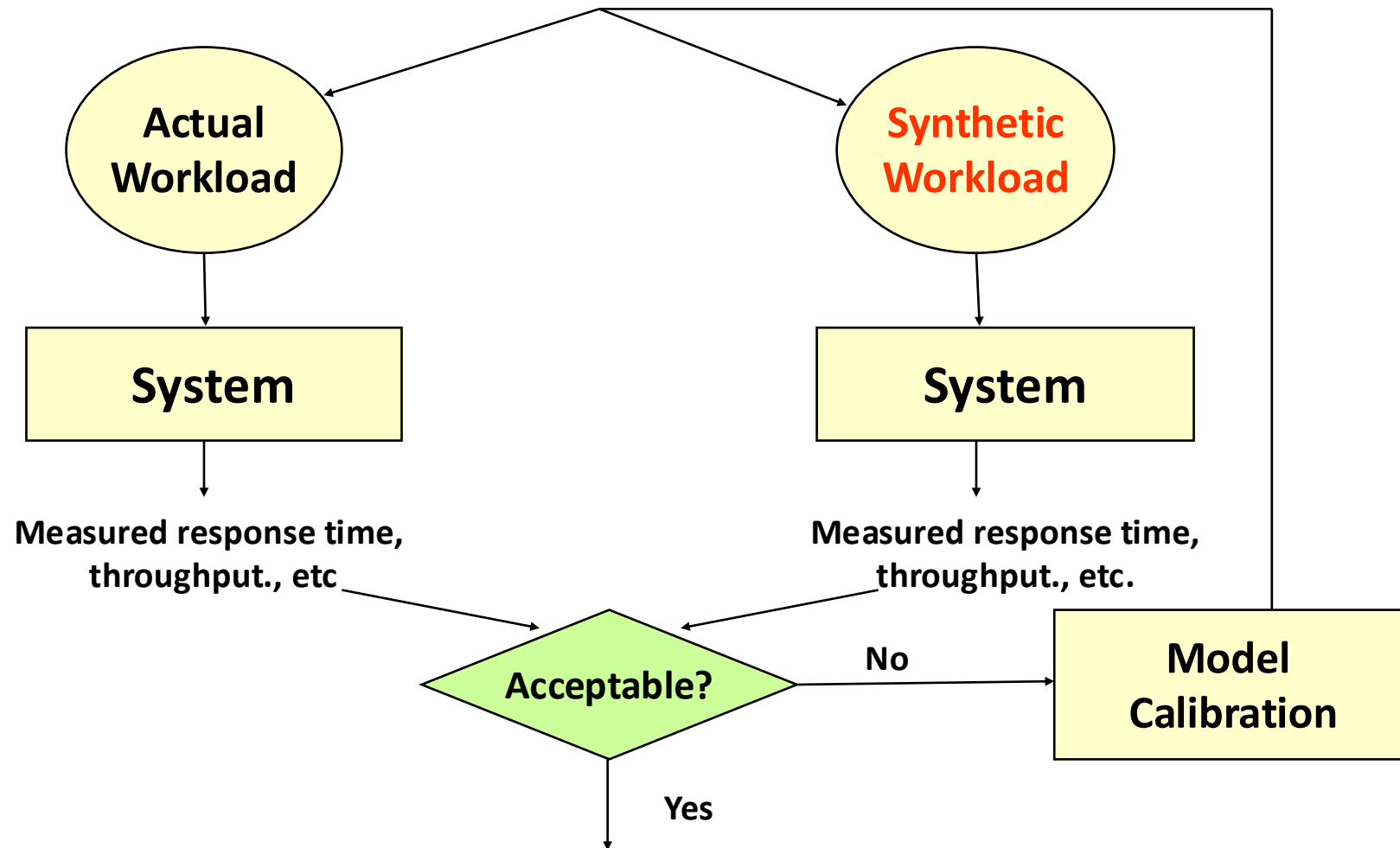
- Describes system's global workload in terms of its main components
- Captures **workload intensity** and **service demand** of each workload component at each resource in the system

Data Collection Issues

- How to determine the parameter values for each basic component?



Validating Workload Models



Workload Forecasting (1)

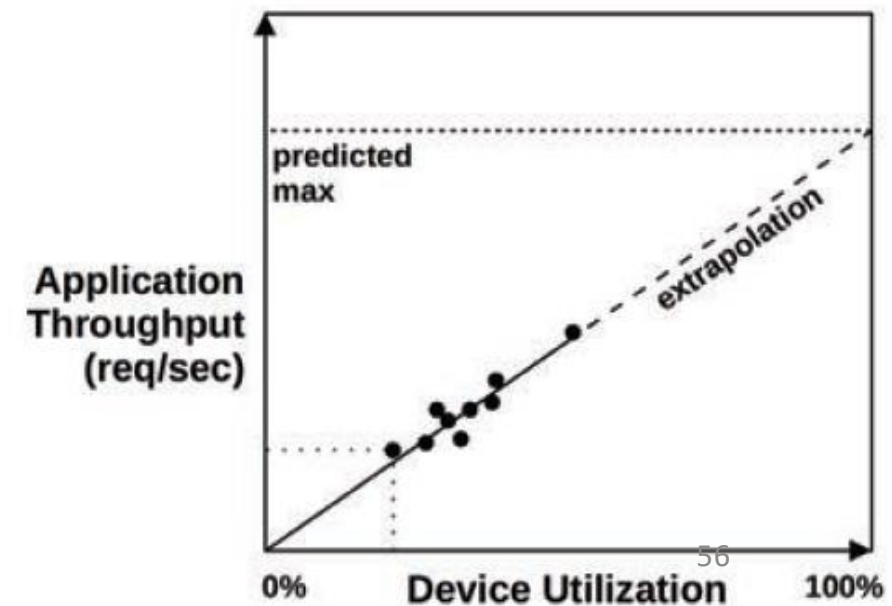
- Predicts workload intensity
 - How will the number of e-mail messages handled per day by the server vary over the next 6 months?
 - How will the number of hits to the corporate intranet's web server vary over time?

Workload Forecasting (2)

- Requires:
 - Evaluate organization's workload trends
 - Analyze historical usage data
 - Analyze business or strategic plans
 - Map plans into business processes (e.g., paperwork reduction will add 50% more e-mail)
- Workload forecasting techniques: moving averages, exponential smoothing, etc.

Resource Limits

- Search for the resource that will become the bottleneck under load
 1. Measure the rate of server requests and monitor this rate over time.
 2. Measure hardware and software resource usage. Monitor this rate over time.
 3. Express server requests in terms of resources used.
 4. Extrapolate server requests to known (or experimentally determined) limits for each resource.



Example: Using Industry Benchmark to Scale Resource Usage

- For a given application, service demand at the server was 10 msec and was obtained in a controlled environment with a server with a SPECint rating of 3.11
- What would be the service demand if the server used in the actual system were faster and had a SPECint rating of 10.4?

ActualServiceDemand = MeasuredServiceDemand x ScalingFactor

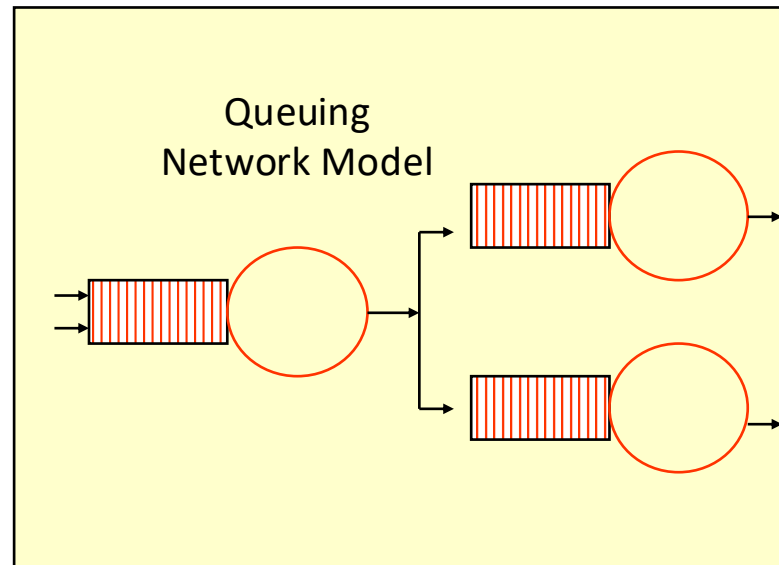
ScalingFactor = ControlledResourceThroughput / ActualResourceThroughput

*ActualServiceDemand = 10 * (3.11/10.4) = 3.0 msec*

Estimating Performance Measures

System Description

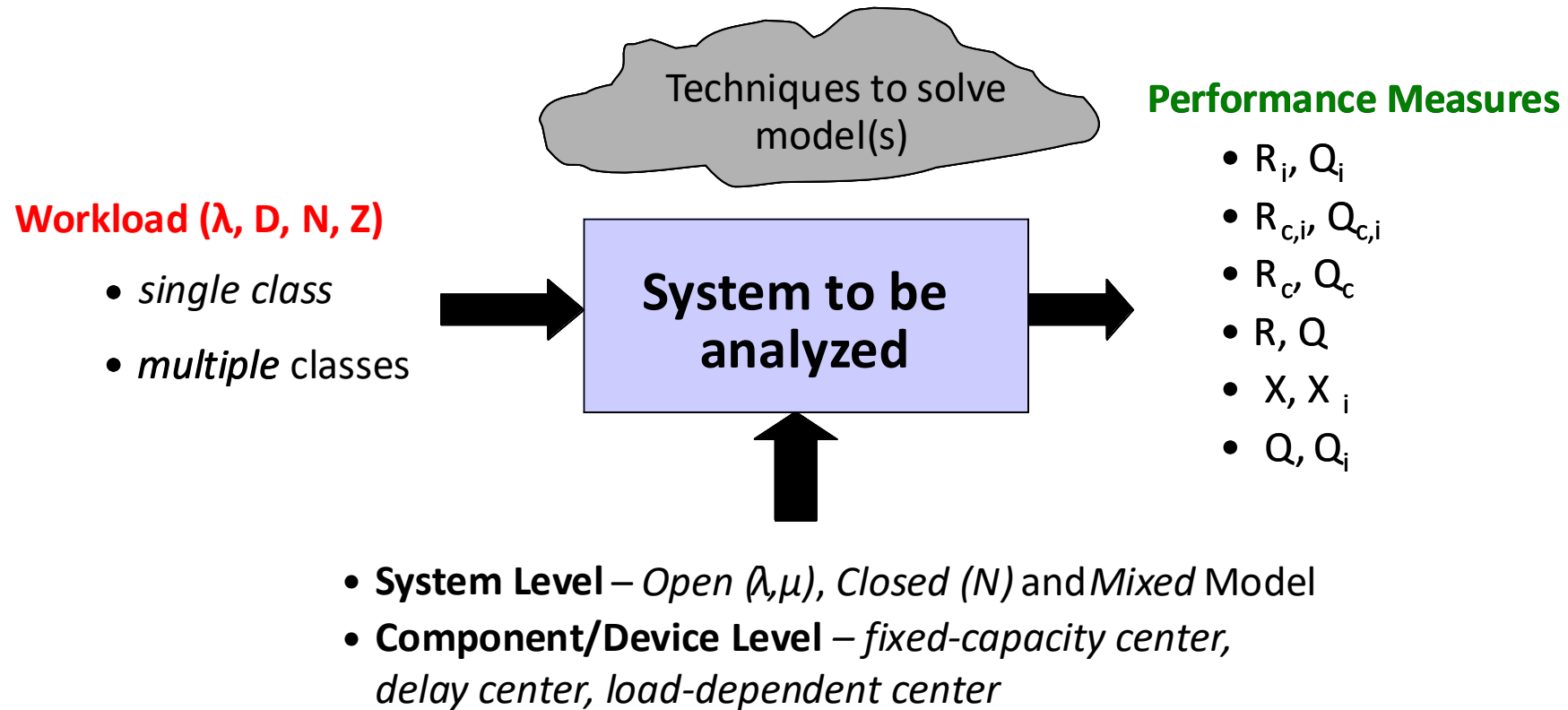
- System parameters
- Resources parameters
- Workload parameters
 - service demands
 - workload intensity



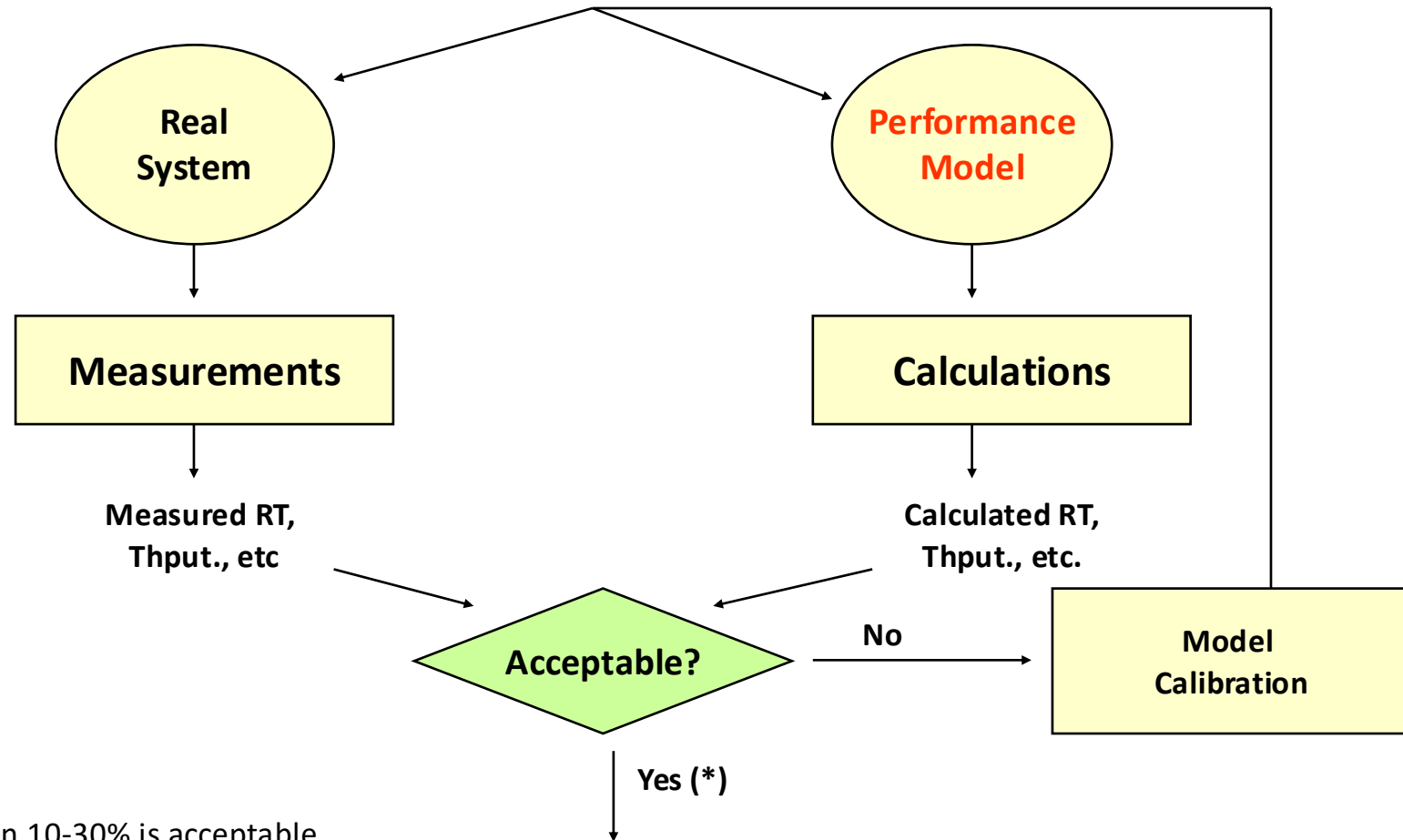
Performance Measures

- response time
- throughput
- utilization
- queue length

Performance Modeling



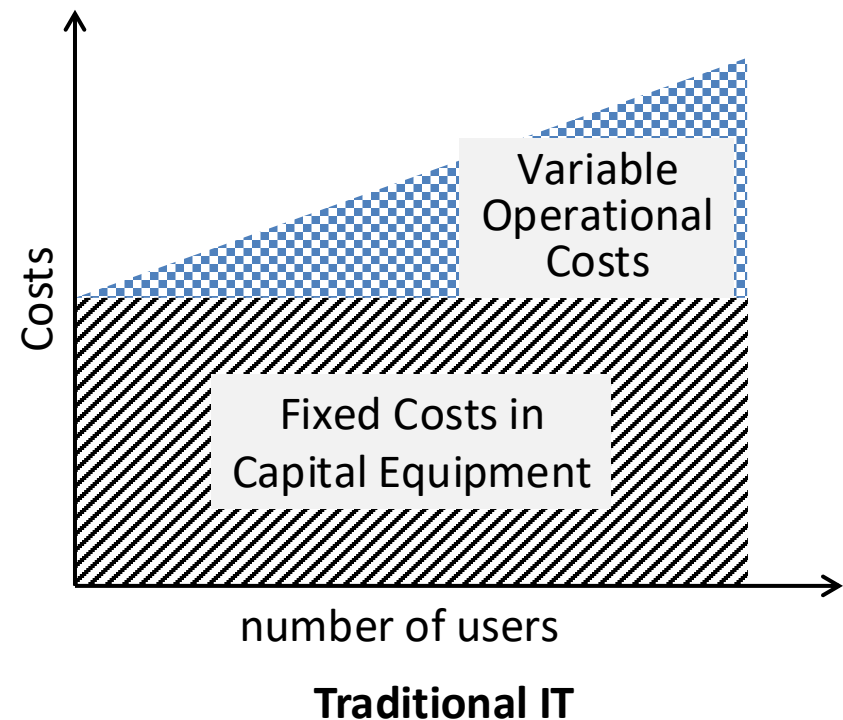
Validating Performance Models



(*) error of between 10-30% is acceptable

Cost Model

- Identifies major sources of cost and determine how cost vary with system size and architecture
 - startup (fixed) costs
 - operating costs



Cost Model: Categories

- Hardware costs: client and server machines, disks, routers, bridges, cabling, maintenance, etc.
- Software costs: operating systems, middleware, DBMS, mail processing software, office automation, applications, etc.
- Telecommunication and electricity costs: WAN services, ISP, etc.
- Support costs: salaries and benefits of all system administrators, help desk support, network people, etc

Summary

- Performance evaluation can be conducted using various methodologies
- Capacity planning is important to ensure that the system can perform when the workload increases
- References:
 - [Gregg20] – Chapter 2